A Comparative Analysis of Deep Learning Models: YOLOv8 vs RT-DETR for Road Damage Detection

Awwab Samad

The University of Texas at Austin

April 20, 2026

© 2026 Awwab Samad. All rights reserved.

Contents

1 Introduction **2**

1.1 Research Objectives 3

2 Materials and Data **4**

2.1 Dataset Description 4

2.2 Data Preprocessing 5

3 Methods **6**

3.1 YOLOv8: A Pure CNN-based Model 6

3.1.1 Architectural Design 6

3.1.2 Objective Loss Functions 7

3.2 RT-DETR: A Hybrid CNN-Transformer Model 8

3.2.1 Architectural Design 9

3.2.2 Objective Loss Functions 10

3.3 Training YOLOv8 and RT-DETR 11

4 Results **11**

4.1 Evaluation Metrics 11

4.2 Training and Validation 12

4.3 Overall Model Performance 13

4.4 Class-Specific Analysis 13

4.5 Error Analysis and Confusion Matrices 14

4.6 Testing and Predictions 15

5 Discussion **15**

5.1 Architectural Differences and Spatial Awareness 17

5.2 Limitations and Future Work 17

6 Conclusion **18**

Abstract

Automated road damage detection is a critical component of modern infrastructure maintenance and transportation safety. While deep learning models offer efficient ways to monitor pavement conditions, selecting the right architecture can be challenging. This study presents a comparative analysis of YOLOv8, a leading convolutional neural network (CNN), and RT-DETR, a hybrid CNN-transformer model, for road damage detection. Both architectures were trained and evaluated on a subset of the RDD-2022 dataset across four distinct damage categories. The results indicate that while both models effectively identified common linear cracks, they ultimately struggled with the class imbalance of minority categories. Overall, RT-DETR achieved a slightly higher overall Mean Average Precision (mAP@0.50) of 0.548 compared to YOLOv8's 0.522. Specifically, RT-DETR's global self-attention mechanisms helped mitigate the spatial limitations of localized convolutions, yielding roughly a 60% relative performance improvement on rare anomalies like potholes, at the cost of higher resource consumption. Although detection rates for minority classes remained low for both models, these findings demonstrate that hybrid CNN-transformers provide a more spatially aware and robust baseline for automated road detection systems.

1 Introduction

The structural health of road surfaces is essential for urban mobility and public safety. When pavements deteriorate, the physical and financial costs are substantial. Structural defects like cracks and potholes increase vehicle wear-and-tear, disrupt traffic flow, and can create severe safety hazards to drivers. In the United States, the American Automobile Association (AAA) reported in 2022 that pothole-related damage cost drivers a staggering $26.5 billion in vehicle repairs in the previous year (American Automobile Association, 2022). Furthermore, poor pavement frequently causes drivers to swerve or lose control, contributing to roadway departures that account for over half of the 36,640 annual U.S. traffic fatalities in 2025 (National Highway Traffic Safety Administration, 2026; U.S. Department of Transportation, 2024). Unfortunately, in the United States, decades of underinvestment and escalating traffic demands have severely deteriorated pavement conditions. In the 2025 Report Card for America's Infrastructure, the American Society of Civil Engineers (ASCE) assigned the nation's infrastructure an overall "C" grade (American Society of Civil Engineers, 2025). Moreover, a 2026 analysis by The Pew Charitable Trusts identified an $86 billion state funding shortfall over the next decade strictly for maintaining existing roads and bridges (The Pew Charitable Trusts, 2026). Traditionally, municipalities have relied on manual visual inspections to find and patch these defects. However, this approach is slow, expensive,

and heavily subjective (Radopoulou & Brilakis, 2016). Because cities cannot feasibly employ enough inspectors to manually monitor every street, there is an urgent need for automated, high-accuracy Road Damage Detection (RDD) systems to catch minor cracks early before they expand into hazardous, costly potholes.

The advent of computer vision and deep learning has revolutionized infrastructure monitoring (Koch et al., 2015). Convolutional Neural Networks (CNNs), particularly the You Only Look Once (YOLO) family, have become the industry standard for real-time object detection because they are fast and efficient (Redmon et al., 2016). However, building a reliable road damage detection model is difficult because real-world road damage data is naturally imbalanced. A standard stretch of road might contain thousands of minor linear cracks but only a handful of severe, isolated failures like potholes (Maeda et al., 2018). Traditional CNNs often struggle to correctly identify these rare anomalies because they only look at small, localized sections of an image at a time. Recently, transformer-based architectures like vision transformers (ViTs) and specifically, hybrid CNN-transformer models like RT-DETR have emerged, offering a new approach by processing the entire image context at once (Vaswani et al., 2017). To evaluate whether this newer architecture performs better for real-world pavement monitoring, this study uses a United States subset of the RDD-2022 dataset to compare YOLOv8 and RT-DETR across four specified damage categories: Longitudinal Cracks, Transverse Cracks, Alligator Cracks, and Potholes. Achieving high precision across all these classes is essential for deploying a reliable damage detection system.

1.1 Research Objectives

The primary objective of this research is to conduct a comparative performance evaluation of a purely convolutional architecture (YOLOv8) and a hybrid CNN-transformer (RT-DETR). The study seeks to address the following research questions:

- **RQ1:** How does the performance of a CNN-based model (YOLOv8) compare to a hybrid CNN-transformer model (RT-DETR) on automated road damage detection?
- **RQ2:** How do the architectural differences between CNNs and hybrid CNN-transformers influence the detection of specific damage geometries, such as linear cracks versus circular potholes?
- **RQ3:** How does an imbalanced dataset with limited minority-class examples affect the detection accuracy of both models?

The remainder of this paper is organized as follows: Section 2 details the materials and data sources, including the specific class distributions of the dataset. Section 3 outlines the methods used, comparing

the underlying architectures and objective functions of YOLO and RT-DETR. Section 4 presents the experimental results and visual comparisons. Section 5 discusses the implications of these findings alongside future work, and Section 6 concludes the study.

2 Materials and Data

The primary data source for this research is a subset of the Road Damage Detection 2022 (RDD-2022) dataset, specifically curated for the United States and hosted on the Roboflow platform (Arya et al., 2024; segmentpavementdiseases, n.d.). This dataset provides the ground-truth labels required to evaluate the performance of object detection models in road damage detection as illustrated on the Dataset Ninja website (Ninja, 2026).

2.1 Dataset Description

The dataset consists of 4,913 high-resolution 640×640 images of different road surfaces captured via vehicle-mounted cameras. It comprises four classes of road cracks: Longitudinal Cracks (D00), Transverse Cracks (D10), Alligator Cracks (D20), and Potholes (D40), as shown in Figure 1. These four categories follow the Federal Highway Administration (FHWA) classification standards for pavement distress (Miller & Bellinger, 2014). The alphanumeric codes (D00–D40) are the standard labels for the road damage classes in the dataset.

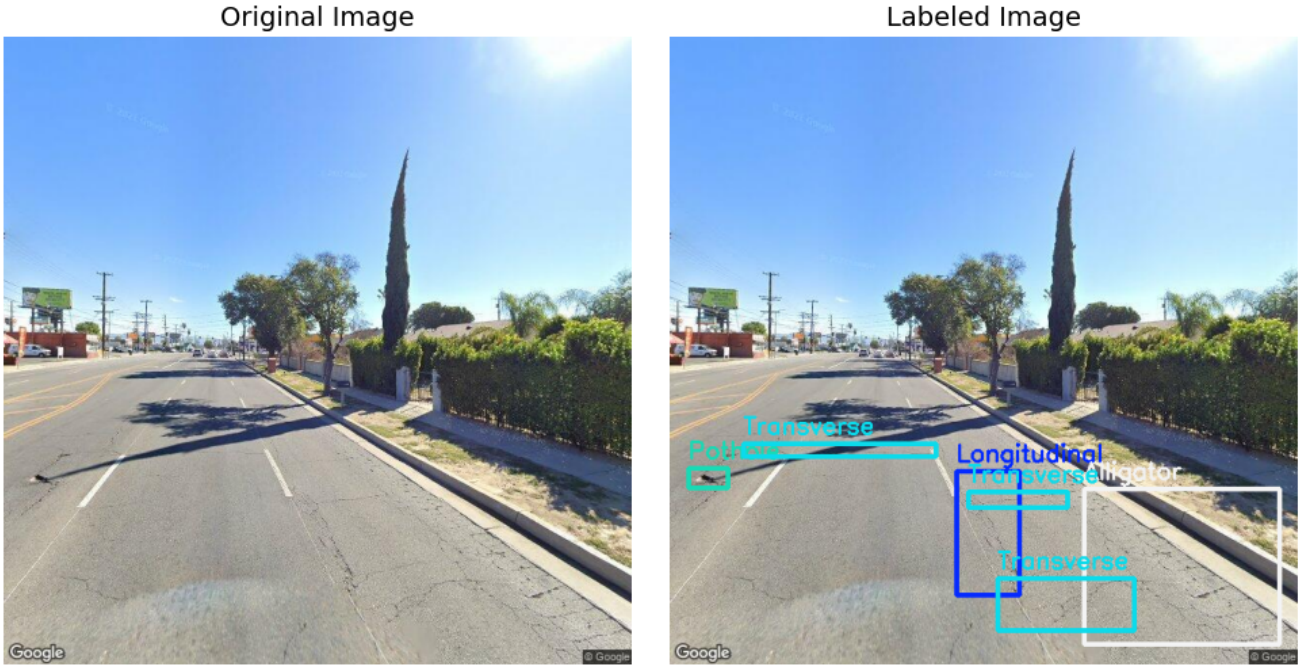


Figure 1: A sample from the dataset showing the original pavement image (left) with its annotated labels (right) of the four damage categories.

Below is a detailed description of each damage class, including its physical characteristics and maintenance priority:

- **Longitudinal Cracks (D00):** Cracks running parallel to the direction of travel, often as a result of poor longitudinal joint construction (Pavement Interactive, n.d.). These are typically lower priority but require sealing to prevent future moisture infiltration.
- **Transverse Cracks (D10):** Cracks running perpendicular to the road center. These typically develop during cold-temperature cycling when the asphalt concrete contracts and fractures under tensile stress (Behnia et al., 2014).
- **Alligator Cracks (D20):** Interconnected cracks forming a pattern resembling the skin of an alligator. This is commonly caused by repeated vehicle loads over a weak sub-grade layer and indicates a significant structural fault (Mohammed et al., 2025). These require a high-priority repair to prevent total road collapse.
- **Potholes (D40):** Bowl-shaped depressions in the pavement surface resulting from water infiltration through existing cracks. They grow rapidly as traffic breaks apart the weakened asphalt (Jassal, 1998), creating an immediate safety hazard for drivers.

Table 1: Class Distribution of the Dataset

Damage Type	Class ID	Instances
Longitudinal Crack	D00	6,883
Transverse Crack	D10	3,384
Alligator Crack	D20	859
Pothole	D40	138

A key challenge in our dataset is the imbalanced class distribution. As shown in Table 1, there is a long-tail pattern in the dataset where longitudinal and transverse cracks significantly outnumber potholes. This is a reflection of the fact that cracks are far more common than potholes in the real world. Potholes are end-stage failures that evolve from initial cracks (Usanga, 2021) so they remain naturally scarce.

2.2 Data Preprocessing

Preprocessing steps were applied to the dataset to ensure uniform input for the deep learning models. These steps included:

- **Resizing:** Images were standardized to a resolution of 640×640 pixels directly from Roboflow, aligning with the requirements of the YOLOv8 and RT-DETR architectures.

- **Augmentation:** In order to mitigate the impact of class imbalance, data augmentation techniques were applied such as mosaic augmentation and random flipping within the YOLOv8 and RT-DETR training loops (Jocher, 2025).

Finally, consistent with standard machine learning practices, the dataset was partitioned into three sets: a training set of 3,435 images (70%), a validation set of 985 images (20%), and a test set of 493 images (10%).

3 Methods

The core methodology of this research involves a side-by-side performance evaluation of two state-of-the-art deep learning models for road damage detection: YOLOv8 (You Only Look Once) and RT-DETR (Real-Time Detection Transformer). We will present a high-level overview of the architectures and loss functions of both these models in this section.

3.1 YOLOv8: A Pure CNN-based Model

YOLOv8 is the eighth iteration of the You Only Look Once (YOLO) model developed by Ultralytics that utilizes a convolutional neural network (CNN) to perform object detection. Unlike earlier YOLO versions that relied on predefined anchor boxes, YOLOv8 utilizes an anchor-free, decoupled head to predict object locations and classes independently, significantly improving detection accuracy and feature extraction efficiency (Jocher et al., 2023).

3.1.1 Architectural Design

The YOLOv8 architecture can be divided into three primary components: the backbone, the neck, and the head (Torres, 2024). Figure 2 illustrates a simplified architecture of the YOLOv8.

- **Backbone:** This is responsible for feature extraction. It uses Cross-Stage Partial with 2 Convolutions (C2f) to improve gradient flow and reduce computational load. It also includes a Spatial Pyramid Pooling - Fast (SPPF) layer at the end of the backbone to capture multi-scale contextual information before passing the features to the neck (Colin, 2024).

- **Neck:** This part uses PANet (Path Aggregation Network) to combine the features extracted by the backbone at different scales. It uses a top-down and bottom-up approach to thoroughly aggregate the features, ensuring that the model can detect cracks of various sizes (Colin, 2024).

- **Head:** The decoupled head separates the classification and bounding box regression tasks into two distinct, parallel branches. It processes the combined features generated by the backbone and neck to predict bounding boxes for the cracks, classify damage categories, and assign confidence scores without relying on predefined anchor boxes (Colin, 2024; Torres, 2024).

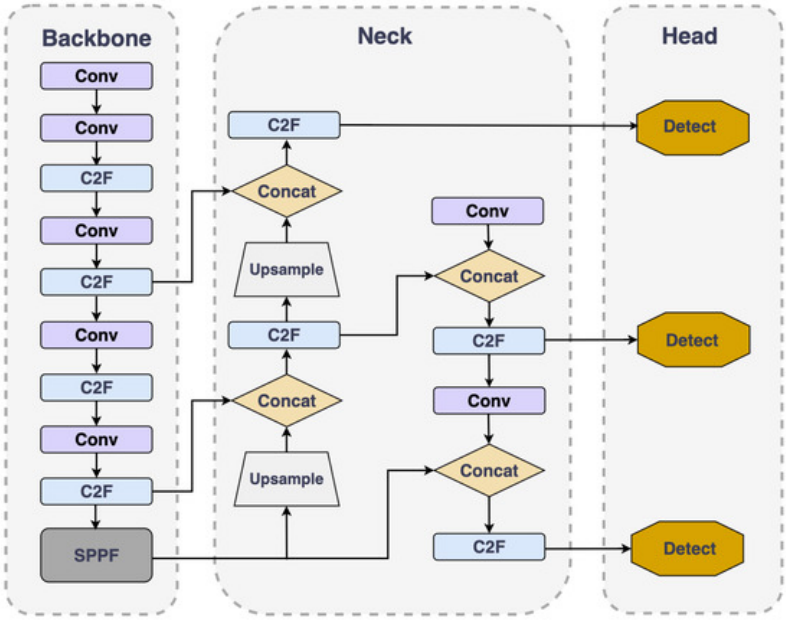


Figure 2: Simplified architecture of the YOLOv8 model (Yao et al., 2024).

3.1.2 Objective Loss Functions

To optimize model weights during training, YOLOv8 uses a composite loss function that calculates the error between the model's predictions and the ground-truth annotations. As standard in the YOLOv8 architecture, the total loss is a combination of three components:

1. **Classification Loss (cls_loss):** Uses the Binary Cross-Entropy (BCE) to determine the probability of the correct damage category (e.g., distinguishing a longitudinal crack from a transverse crack). The formula is expressed as:

$$L_{cls} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)]$$

   Where N is the total number of samples, y_i represents the true class label (0 or 1), and $\hat{p}_i$ is the predicted probability of the class (GeeksforGeeks, 2025).

2. **Bounding Box Loss (box_loss):** Calculates Complete Intersection over Union ($CIoU$) to penalize inaccuracies in predicted bounding box coordinates. This metric improves upon the standard

Intersection over Union (IoU), which is defined as the ratio of the intersection area to the union area of the predicted box A and ground truth B (V7, 2022):

$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

While IoU only measures overlap, $CIoU$ also includes the normalized distance between center points and an aspect ratio consistency term. The loss function is expressed as:

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v$$

where ρ is the Euclidean distance between the center points b and b^{gt}, c is the diagonal length of the smallest enclosing box covering both boxes, and v measures aspect ratio consistency (Zheng et al., 2020).

3. **Distribution Focal Loss (*dfl_loss*):** Instead of treating bounding box coordinates as absolute values, DFL models the coordinates as a continuous probability distribution (Li et al., 2020). This is useful for road damage detection when the exact edges of cracks are unclear. The formula is expressed as:

$$L_{DFL}(S_i, S_{i+1}) = -\left((y_{i+1} - y)\log(S_i) + (y - y_i)\log(S_{i+1})\right)$$

Where y is the continuous coordinate label, y_i and y_{i+1} are the two nearest discrete values, and S_i and S_{i+1} are the predicted probabilities for those discrete values.

Thus, the total loss is calculated as follows:

$$L_{total} = \lambda_{cls} L_{cls} + \lambda_{box} L_{CIoU} + \lambda_{dfl} L_{DFL}$$

Where λ_{cls}, λ_{box}, and λ_{dfl} are the respective hyperparameter weights assigned to each loss component.

3.2 RT-DETR: A Hybrid CNN-Transformer Model

RT-DETR, Baidu's Real-Time Detection Transformer, is a cutting-edge end-to-end object detector that provides real-time performance while maintaining high accuracy. It was the first transformer-based detector to achieve performance comparable to the YOLO series while leveraging the power of global self-attention and eliminating the need for Non-Maximum Suppression (NMS) (Zhao et al., 2023).

3.2.1 Architectural Design

Unlike the purely convolutional YOLOv8, RT-DETR replaces the traditional CNN neck and head with a hybrid architecture that incorporates transformer blocks into an encoder and decoder. It consists of four main components, as shown in Figure 3: the backbone, the hybrid encoder, the query selector, and the transformer decoder (Reddy, 2024).

- **Backbone:** Similar to YOLO, RT-DETR utilizes a CNN backbone (such as ResNet or HGNet) to extract initial multi-scale image features from the road images (Reddy, 2024).
- **Hybrid Encoder:** The encoder processes these multiscale features using two key modules: an Attention-based Intra-scale Feature Interaction (AIFI) module to process high-level features on the same scale, and a Cross-Scale Feature-Fusion Module (CCFM) to fuse features across different scales. This decoupled design reduces computational costs while preserving the high-resolution details necessary for detecting thin cracks (Jocher, 2023b; Reddy, 2024).
- **Uncertainty-minimal Query Selection:** This module selects the top-K highest-quality image features to serve as initial object queries for the decoder (Reddy, 2024).
- **Transformer Decoder & Head:** The decoder iteratively refines the selected queries by integrating spatial information with the image features from the encoder through self-attention mechanisms. This allows the model to globally reason about the relationship between distant pixels (such as a crack and its surrounding background). Bounding box coordinates and confidence scores are then generated (Reddy, 2024).

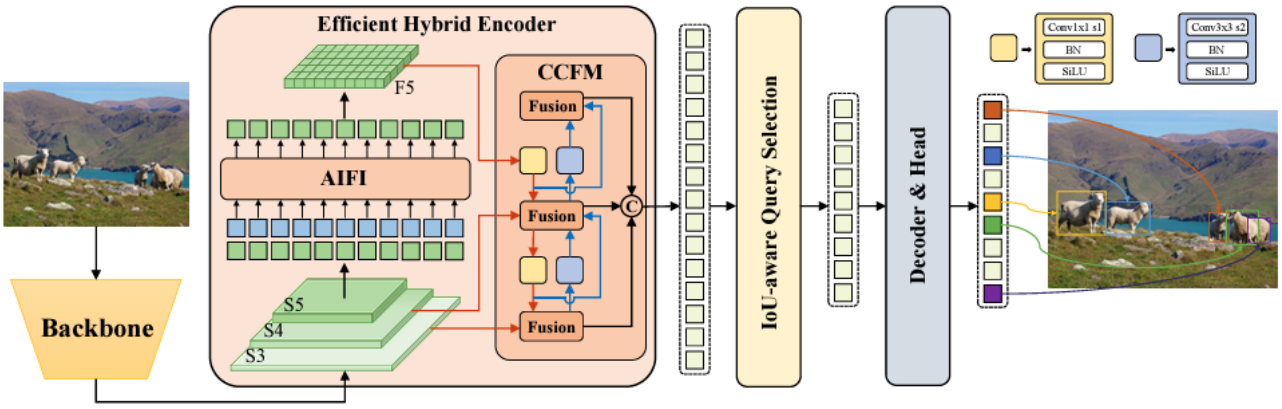


Figure 3: Architecture of the RT-DETR model (Zhao et al., 2023).

3.2.2 Objective Loss Functions

Unlike YOLOv8, RT-DETR does not rely on Non-Maximum Suppression (NMS) to filter duplicate detections. Instead, it uses bipartite matching via the Hungarian algorithm to ensure each object is predicted only once (Zhao et al., 2023). Therefore, its loss functions differ from YOLOv8 so that the transformer decoder can accurately predict coordinates and classes in a single step. The total loss is a weighted sum of three components:

1. **Classification Loss (cls_loss):** Uses the Varifocal Loss (VFL), an IoU-aware extension of Focal Loss, to penalize incorrect class predictions. While standard Focal Loss treats all positive samples equally, VFL weights positive samples by their localization accuracy. The formula is expressed as:

$$L_{cls} = \begin{cases} -q\left(q\log(p) + (1-q)\log(1-p)\right) & \text{if } q > 0 \\ -\alpha p^{\gamma}\log(1-p) & \text{if } q = 0 \end{cases}$$

   Where p is the predicted class score, q is the IoU target score (which is > 0 for positive samples and 0 for negative background samples), α balances the loss, and γ is the focusing parameter (Zhang et al., 2021).

2. L_1 **Loss ($l1_loss$):** Calculates the Mean Absolute Error (MAE) between the predicted bounding box coordinates and the ground truth, providing a stable, unscaled gradient during early training epochs. The formula is expressed as:

$$L_1 = \sum_{i \in \{x,y,w,h\}} |b_i - b_i^{gt}|$$

   Where b_i and b_i^{gt} are vectors representing the center coordinates, width, and height of the predicted and ground-truth boxes, respectively (Carion et al., 2020).

3. **Generalized IoU Loss ($giou_loss$):** While standard Intersection over Union (IoU) fails to provide gradients when boxes do not overlap, Generalized IoU (GIoU) solves this by factoring in the smallest enclosing box. This is crucial for detecting sparse, isolated anomalies like potholes (Rezatofighi et al., 2019). The formula is expressed as:

$$L_{GIoU} = 1 - IoU + \frac{|C \setminus (A \cup B)|}{|C|}$$

   Where A and B are the predicted and ground truth bounding boxes, and C represents the smallest convex shape enclosing both A and B.

Thus, the total loss is calculated as follows:

$$L_{total} = \lambda_{cls} L_{cls} + \lambda_{l1} L_1 + \lambda_{giou} L_{GIoU}$$

Where λ_{cls}, λ_{l1}, and λ_{giou} are the respective hyperparameter weights assigned to each loss component.

3.3 Training YOLOv8 and RT-DETR

Training was conducted using the Ultralytics framework (Jocher, 2023a; Skalski, 2024) on an NVIDIA T4 GPU provided by Google Colab. Table 2 shows the parameters that were utilized in the training of both models. The optimizer, momentum, and learning rate were set automatically by the Ultralytics framework while the training epochs, batch size, augment, and patience parameters were set manually. The total training time was approximately 2 hours for YOLOv8 and 6 hours for RT-DETR.

Table 2: Training Parameters used for YOLOv8 and RT-DETR

Parameter	**Value**
Input Resolution	640×640
Training Epochs	100
Batch Size	16
Augment	True
Optimizer	AdamW
Patience	15
Momentum	0.9
Learning Rate	0.001

4 Results

This section presents the results of the YOLOv8 and RT-DETR models. The evaluation was conducted using the test subset of 493 images. The following subsections define the evaluation metrics, analyze the training process, and provide a breakdown of both overall and class-specific performance. Finally, an error analysis is conducted via confusion matrices and qualitative visual comparisons to identify the strengths and weaknesses of each architecture.

4.1 Evaluation Metrics

Model performance was evaluated using Precision, Recall, and Mean Average Precision (mAP). Precision measures the accuracy of the positive predictions, while Recall measures the model's ability to find

all actual positive instances (McDonough, 2024). They are calculated using True Positives (TP), False Positives (FP), and False Negatives (FN):

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

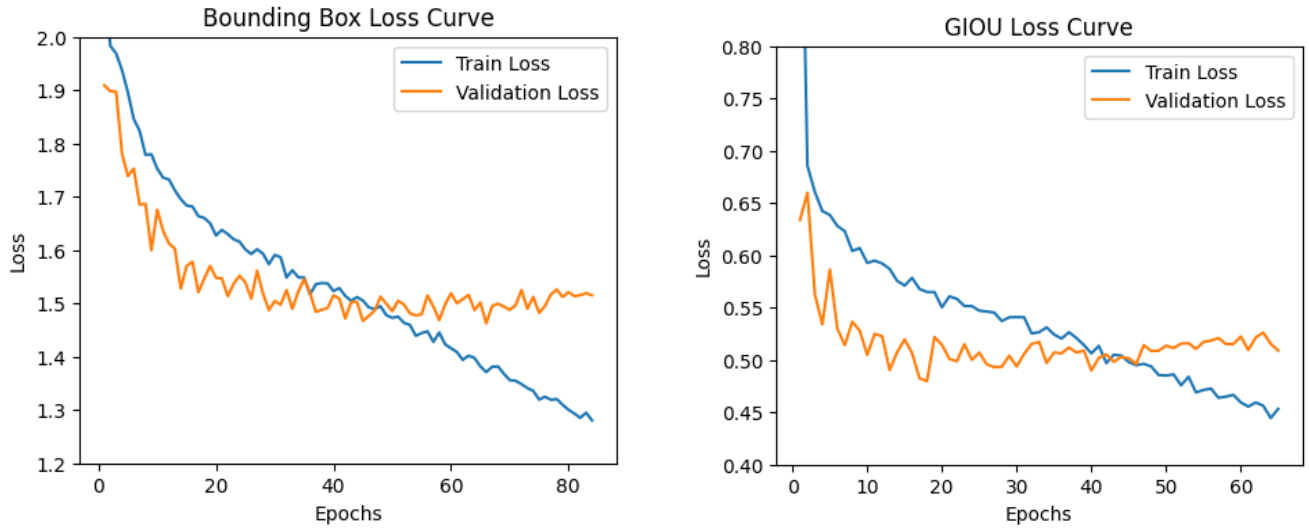
To provide a comprehensive performance score, Average Precision (AP) calculates the area under the Precision-Recall curve for a single class. The Mean Average Precision (mAP) is then calculated by taking the average of the AP scores across all N classes in the dataset (V7, 2022):

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i$$

For this study, mAP was evaluated at an Intersection over Union (IoU) threshold of 0.50 (mAP@0.50), which is a common benchmark for identifying structural road damage. Finally, Inference Speed, the time taken to process a single frame in milliseconds, was recorded to evaluate the real-time deployment viability of each architecture.

4.2 Training and Validation

Prior to evaluating the final testing metrics, the training and validation loss curves were analyzed to verify model convergence and diagnose potential overfitting. Figure 4 illustrates the Bounding Box Loss for YOLOv8 and the Generalized IoU (GIoU) Loss for RT-DETR.



(a) YOLOv8 Bounding Box Loss

(b) RT-DETR GIoU Loss

Figure 4: Training and validation loss curves for YOLOv8 and RT-DETR.

For both architectures, the training and validation losses showed a sharp initial decline before transitioning into a steadier, gradual decrease. While the training loss continued to improve slowly, the validation curves began to flatten, implying that the models had reached their optimal learning capacity. The validation curves generally followed the training curves closely without any significant upward divergence. This indicates that both models generalized well to unseen data and did not suffer from severe overfitting during the training phase. Due to the early stopping patience parameter (set to 15), YOLOv8 stopped training at epoch 85, while RT-DETR converged more quickly, stopping at epoch 65.

4.3 Overall Model Performance

Both architectures demonstrated high competency in identifying the most frequent damage classes: longitudinal (D00) and transverse (D10). However, the overall metrics reveal distinct behavioral differences between the CNN and the hybrid CNN-transformer.

Table 3: Overall Performance Comparison of YOLOv8 and RT-DETR on the Test Set

Model	Precision	Recall	mAP@0.50	Inference Speed (ms)
YOLOv8	**0.566**	0.474	0.522	**13.28**
RT-DETR	0.502	**0.569**	**0.548**	17.38

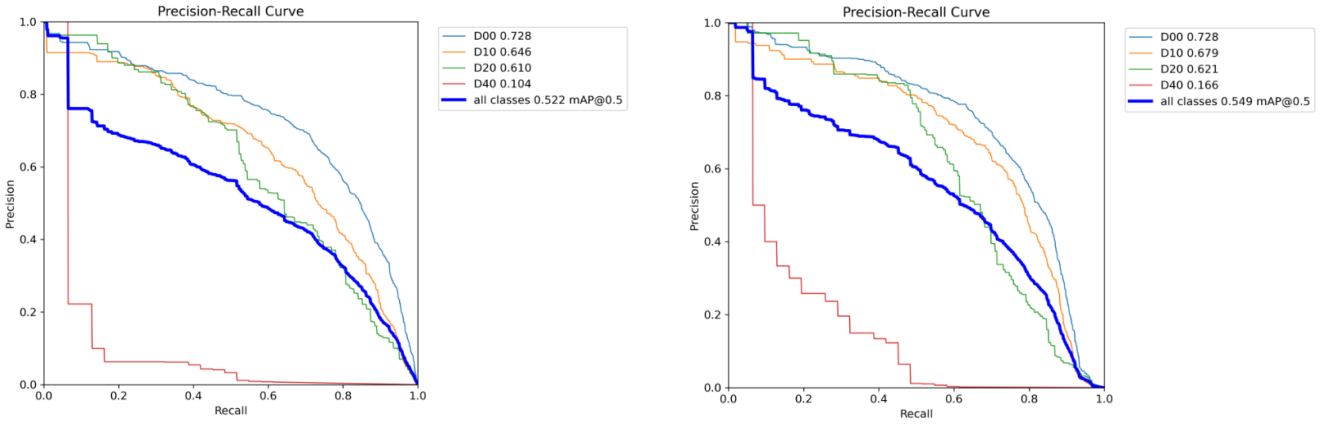
As shown in Table 3, YOLOv8 achieved a higher overall precision, indicating fewer false positives across the dataset. Conversely, RT-DETR demonstrated a significantly higher recall and a higher overall mAP@0.50. However, the inference speed of YOLOv8 was approximately 4 milliseconds faster per frame than that of RT-DETR.

4.4 Class-Specific Analysis

To evaluate model performance for each road damage class, Precision-Recall (PR) curves for both models were generated as shown in Figure 5.

Both models performed exceptionally well on the majority class of longitudinal cracks (D00), with both YOLOv8 and RT-DETR achieving an identical mAP of 0.728. For both transverse cracks (D10) and alligator cracks (D20), RT-DETR demonstrated slight but consistent superiority; it achieved an mAP of 0.679 for Transverse Cracks (D10) and 0.621 for Alligator Cracks (D20), compared to YOLOv8's scores of 0.646 and 0.610, respectively.

However, severe performance degradation is seen for the minority pothole class (D40), which suffers from a severe lack of training examples compared to the other damage classes. For YOLOv8, the PR curve for D40 is severely low, achieving an mAP of only 0.104. RT-DETR demonstrated a slight improvement, increasing the D40 mAP to 0.166. This roughly 60% relative performance increase suggests



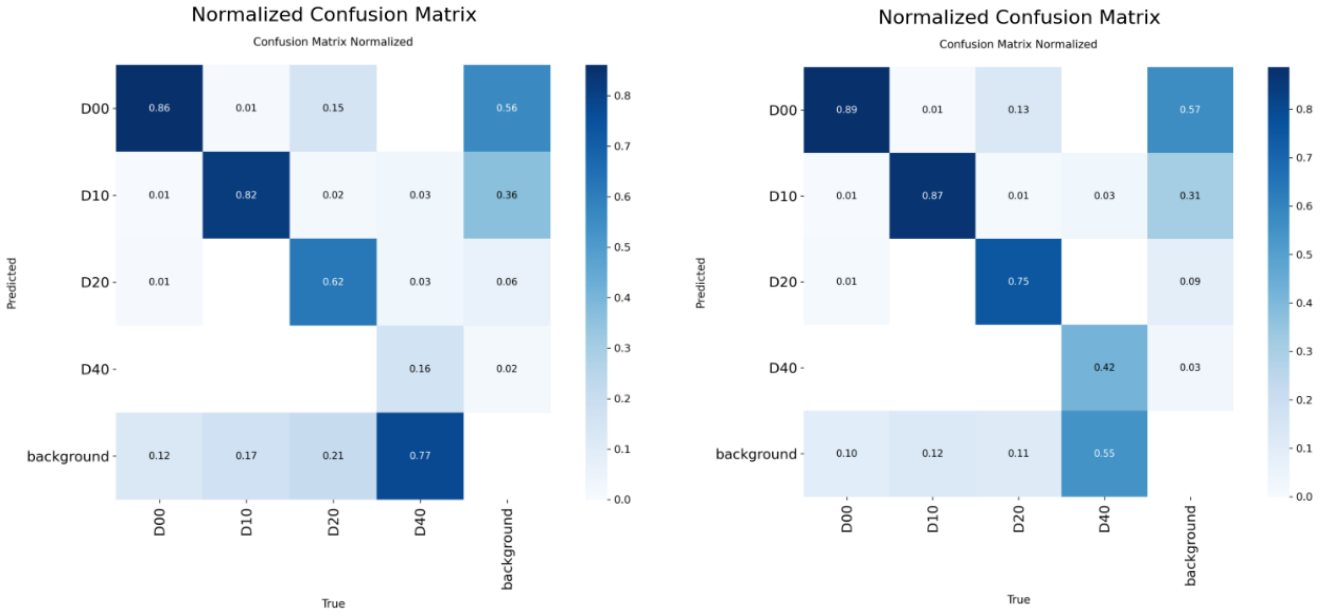
(a) YOLOv8 (b) RT-DETR

Figure 5: Precision-Recall curves for YOLOv8 and RT-DETR.

that the transformer-based architecture is more resilient when detecting isolated anomalies in imbalanced data conditions.

4.5 Error Analysis and Confusion Matrices

An analysis of the normalized confusion matrices in Figure 6 shows the correct predictions and misclassifications of both models trained on the dataset. The confusion matrices were normalized to scale raw detection counts into a range of 0 to 1, allowing for a fair comparison across all classes regardless of their sample size.



(a) YOLOv8 (b) RT-DETR

Figure 6: Normalized confusion matrices for YOLOv8 and RT-DETR.

Looking at the true positive rates on the diagonal axis, RT-DETR outperformed YOLOv8 across every single damage category. YOLOv8 correctly predicted 86% of D00, 82% of D10, 62% of D20, and

a mere 16% of D40 instances. In contrast, RT-DETR correctly predicted 89% of D00, 87% of D10, 75% of D20, and significantly improved pothole detection by successfully identifying 42% of D40 instances.

Regarding misclassifications, the matrices confirm that background hallucination (predicting cracks as normal pavement) is the primary source of error for both models. YOLOv8 exhibited a massive 77% false-negative rate for class D40. While RT-DETR did successfully reduce this background misclassification to 55%, it still remained considerably high. Similar trends are seen in the other classes: YOLOv8 classified 21% of Alligator Cracks (D20) and 17% of Transverse Cracks (D10) with the background, while RT-DETR reduced these specific errors to 11% and 12%, respectively. Although RT-DETR is the objectively stronger model, these matrices highlight the limits of both architectures when applied to highly imbalanced data.

4.6 Testing and Predictions

Finally, after training, both models were tested on real-world road images. Figure 7 presents a direct visual comparison of YOLOv8 and RT-DETR predictions across three images.

Firstly, RT-DETR generally gives a higher confidence score across identical cracks; for example, in the top row, the confidence score for the alligator crack nearly doubles from 0.46 (YOLOv8) to 0.85 (RT-DETR). Secondly, YOLOv8 struggles with complex crack geometries. This is evident in the middle row, where YOLOv8 incorrectly labels an area of alligator cracks as a transverse crack and a large longitudinal crack as a pothole, contributing to false positives. In contrast, RT-DETR correctly classifies the texture as a broad alligator crack (0.93 confidence). Finally, as seen in the bottom row, RT-DETR successfully detects faint transverse and longitudinal cracks that YOLOv8 missed entirely. Furthermore, both models detected the distant pothole in the bottom scene, but RT-DETR did so with significantly higher confidence (0.67 versus YOLOv8's 0.35).

5 Discussion

This section interprets the experimental results to answer the core research questions established in the introduction. Specifically, it examines the overall performance of the two models (RQ1), analyzes the architectural differences influencing detection accuracy (RQ2), and evaluates their response to the severe class imbalance (RQ3).

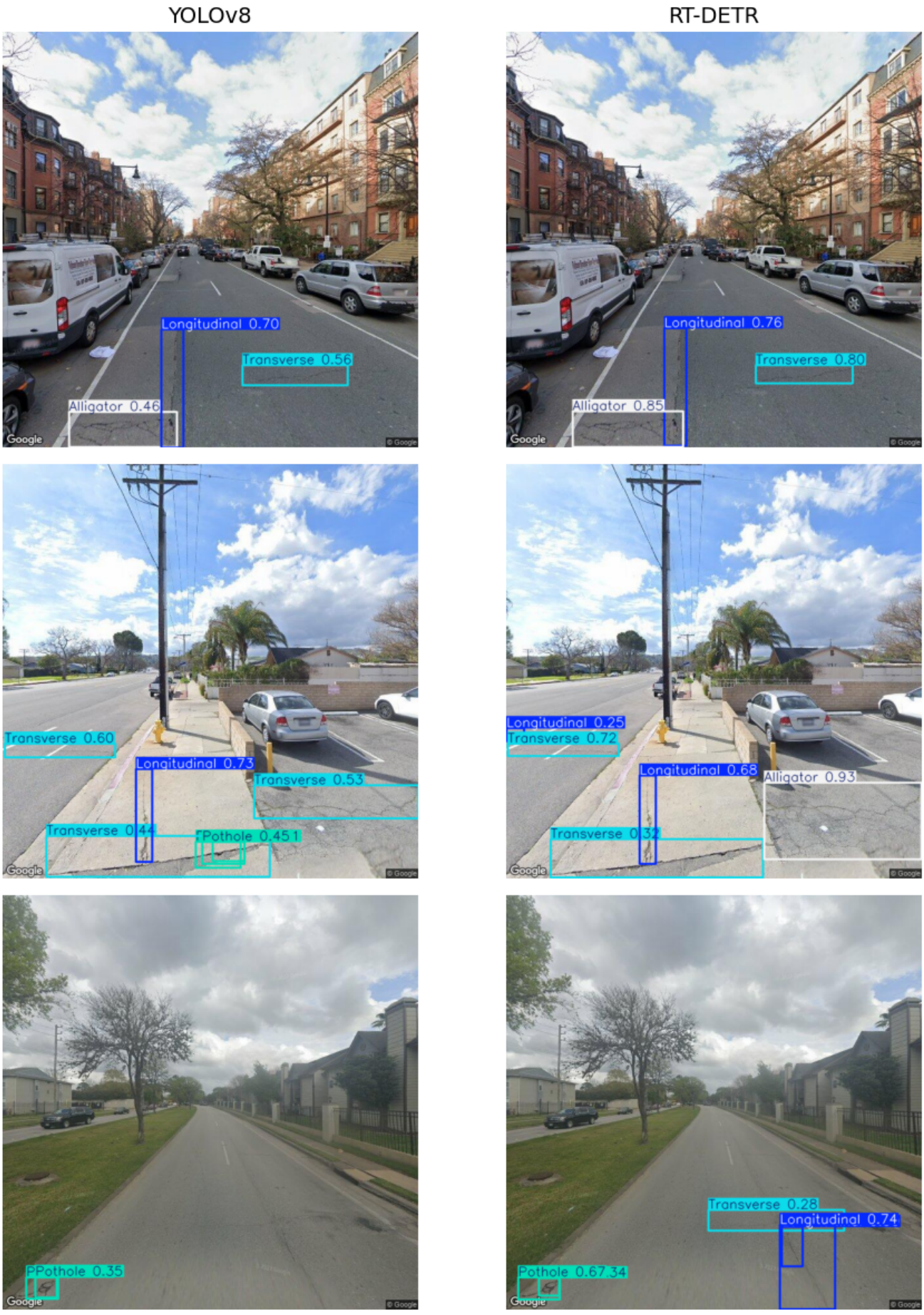


Figure 7: Visual comparison of YOLOv8 (left) and RT-DETR (right) across all four road damage categories.

5.1 Architectural Differences and Spatial Awareness

Addressing RQ1 and RQ2, the results show that YOLOv8 is a fast and capable baseline for general object detection. Its simpler convolutional structure allowed for rapid inference, achieving a processing speed of just 13.28 ms per image during testing, and requiring approximately 2 hours to fully train. However, this purely convolutional structure limited its spatial awareness when dealing with rare, oddly shaped defects. CNNs process images through small, local windows (kernels). This makes it difficult for YOLOv8 to tell the difference between a sparse, geometric feature like a pothole and normal background textures on the road.

Conversely, RT-DETR uses a Hybrid Encoder and self-attention mechanisms to globally analyze the entire image at once (Vaswani et al., 2017). While this architectural overhead resulted in a higher inference latency of 17.38 ms per image and a longer training time of approximately 6 hours, the model yielded more reliable confidence scores and fewer false negatives compared to YOLOv8. Processing the image globally expanded spatial awareness and enabled the model to better distinguish the irregular boundaries of potholes and cracks from complex surrounding textures. Consequently, RT-DETR achieved a roughly 60% relative increase in mAP for the pothole class, as well as improved accuracy for transverse and alligator cracks. These results demonstrate a clear trade-off: although transformer-based architectures require more training time and higher computational resources, they handle complex spatial features and geometries of road distress better than pure CNNs (Jia, 2025).

5.2 Limitations and Future Work

Regarding RQ3, the primary limitation of this study was the severe class imbalance. Due to Google Colab resource limitations, the models were trained on a restricted subset of 4,913 images of the full 47,420-image RDD-2022 dataset. This limited the models' exposure to minority class examples, leaving only 31 validation instances for potholes (D40) compared to 1,389 instances for longitudinal cracks (D00). Consequently, while RT-DETR proved more resilient, its 55% false-negative rate for potholes, alongside persistent background confusion (notably a 57% false-positive rate for longitudinal cracks (D00) being misidentified as background), shows that it still struggled. This proves that while choosing a better architecture is important, it cannot completely replace the need for a balanced, high-volume dataset.

Future work in this domain should therefore focus directly on addressing this class imbalance, most notably scaling up computational resources to train these architectures on bigger datasets. Alternatively, applying data augmentation techniques or generating synthetic data for minority classes like alligator cracks and potholes will be essential to achieve better and more reliable prediction accuracies required

for real-world deployment.

6 Conclusion

In conclusion, automated road damage detection is a necessary step toward smarter and safer urban infrastructure maintenance. This study provided a comparative evaluation of a traditional CNN-based, YOLOv8, and a hybrid CNN-transformer model, RT-DETR. The findings establish that while hybrid CNN-transformers require more computational resources and longer training times, they offer a clear advantage over standard CNNs in spatial awareness and detection capability. By leveraging global attention mechanisms, RT-DETR improved the detection of rarer classes such as potholes without sacrificing the accuracy of common cracks. While data imbalance limited model accuracies, this research demonstrates that moving from local CNN convolutions to global self-attention transformers provides a more robust and spatially aware foundation for the next generation of automated road detection systems.

References

American Automobile Association. (2022, March). Potholes pack a punch as drivers pay $26.5 billion in related vehicle repairs. https://newsroom.aaa.com/2022/03/aaa-potholes-pack-a-punch-as-drivers-pay-26-5-billion-in-related-vehicle-repairs/

American Society of Civil Engineers. (2025, March). *2025 report card for america's infrastructure: A comprehensive assessment of america's infrastructure* (tech. rep.). ASCE. https://infrastructurereportcard.org/wp-content/uploads/2025/03/Full-Report-2025-Natl-IRC-WEB.pdf/

Arya, D., Maeda, H., Ghosh, S. K., Toshniwal, D., & Sekimoto, Y. (2024). Rdd2022: A multi-national image dataset for automatic road damage detection. *Geoscience Data Journal*, *11*, 846–862. https://doi.org/10.1002/gdj3.260

Behnia, B., Buttlar, W. G., & Reis, H. (2014). Cooling cycle effects on low temperature cracking characteristics of asphalt concrete mixture. *Materials and Structures*, *47*(8), 1359–1371. https://doi.org/10.1617/s11527-014-0310-y

Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., & Zagoruyko, S. (2020). End-to-end object detection with transformers. *European Conference on Computer Vision (ECCV)*, 213–229. https://doi.org/10.1007/978-3-030-58452-8_13

Colin, M. (2024, October). *YOLOv8 explained: Understanding object detection from scratch*. https://medium.com/@melissa.colin/yolov8-explained-understanding-object-detection-from-scratch-763479652312

GeeksforGeeks. (2025, July). Binary cross entropy/log loss for binary classification. https://www.geeksforgeeks.org/deep-learning/binary-cross-entropy-log-loss-for-binary-classification/

Jassal, K. S. (1998). *Development of potholes from cracks in flexible pavements* [Master's thesis, Concordia University]. https://www.collectionscanada.gc.ca/obj/s4/f2/dsk2/ftp01/MQ39100.pdf

Jia, C. (2025). Comparison of yolov8 and rt-detr for road damage detection. *2025 International Conference on Power, Electrical Engineering, Electronics and Control (PEEEC)*, 39–43. https://doi.org/10.1109/PEEEC67807.2025.00013

Jocher, G. (2023a, November). Model training with Ultralytics YOLO. https://docs.ultralytics.com/modes/train/

Jocher, G. (2023b, November). Real-time detection transformer (RT-DETR). https://docs.ultralytics.com/models/rtdetr/

Jocher, G. (2025, April). Yolo data augmentation guide. https://docs.ultralytics.com/guides/yolo-data-augmentation/

Jocher, G., Chaurasia, A., & Qiu, J. (2023, November). *Ultralytics yolov8* (Version 8.0.0). GitHub. https://docs.ultralytics.com/

Koch, C., Doycheva, K., Kasi, V., Akinci, B., & Fieguth, P. (2015). A review on computer vision based defect detection and condition assessment of concrete and asphalt civil infrastructure. *Advanced Engineering Informatics*, *29*, 196–210. https://doi.org/10.1016/j.aei.2015.01.008

Li, X., Wang, W., Wu, L., Chen, S., Hu, X., Li, J., Tang, J., & Yang, J. (2020). Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection. *Advances in Neural Information Processing Systems*, *33*, 21002–21012. https://doi.org/10.48550/arXiv.2006.04388

Maeda, H., Sekimoto, Y., Seto, T., Kashiyama, T., & Omata, H. (2018). Road damage detection and classification using deep neural networks with smartphone images: Road damage detection and classification. *Computer-Aided Civil and Infrastructure Engineering*, *33*, 1127–1141. https://doi.org/10.1111/mice.12387

McDonough, M. (2024, November). Precision and recall. https://www.britannica.com/science/precision-and-recall

Miller, J. S., & Bellinger, W. Y. (2014). *Distress identification manual for the long-term pavement performance program* (5th Revised, Report No. FHWA-HRT-13-092). Federal Highway Administration, U.S. Department of Transportation. McLean, VA. https://www.fhwa.dot.gov/publications/research/infrastructure/pavements/ltpp/13092/13092.pdf

Mohammed, A., Teama, Z., & Zaidan, M. (2025). Investigation of the reasons for alligator cracks at side of flexible pavement roads. *Annales de Chimie - Science des Matériaux*, *49*, 37–42. https://doi.org/10.18280/acsm.490105

National Highway Traffic Safety Administration. (2026, April). *Early estimate of motor vehicle traffic fatalities and fatality rate in 2025* (tech. rep.). U.S. Department of Transportation. https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813800

Ninja, D. (2026, April). Visualization tools for rdd2022 dataset. https://datasetninja.com/road-damage-detector

Pavement Interactive. (n.d.). Longitudinal cracking. https://pavementinteractive.org/reference-desk/pavement-management/pavement-distresses/longitudinal-cracking

Radopoulou, S. C., & Brilakis, I. (2016). Automated detection of multiple pavement defects. *Journal of Computing in Civil Engineering*, *31*(2), 04016057. https://doi.org/10.1061/(ASCE)CP.1943-5487.0000623

Reddy, N. (2024, July). RT-DETR: The Next Evolution in Real-Time Object Detection. https://medium.com/@nandinilreddy/rt-detr-the-next-evolution-in-real-time-object-detection-aa1c7880f368

Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You only look once: Unified, real-time object detection. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 779–788. https://doi.org/10.1109/CVPR.2016.91

Rezatofighi, H., Tsoi, N., Gwak, J., Sadeghian, A., Reid, I., & Savarese, S. (2019). Generalized intersection over union: A metric and a loss for bounding box regression. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 658–666. https://doi.org/10.1109/CVPR.2019.00075

segmentpavementdiseases. (n.d.). Rdd2022 computer vision dataset. https://universe.roboflow.com/segmentpavementdiseases/rdd2022-shuvm

Skalski, P. (2024, July). How to train rt-detr on a custom dataset with transformers. https://blog.roboflow.com/train-rt-detr-custom-dataset-transformers/

The Pew Charitable Trusts. (2026, July). *States are falling behind on roadway maintenance* (tech. rep.). The Pew Charitable Trusts. https://www.pew.org/en/research-and-analysis/reports/2025/07/states-fall-short-of-funding-needed-to-keep-roads-and--bridges-in-good-repair

Torres, J. (2024, March). *YOLOv8 architecture explained.* https://yolov8.org/yolov8-architecture-explained/

U.S. Department of Transportation. (2024, February). *2024 national roadway safety strategy progress report* (tech. rep.). U.S. Department of Transportation. https://www.transportation.gov/sites/dot.gov/files/2024-02/2024%20NRSS%20Progress%20Report.pdf

Usanga, I. (2021). Comparative analysis of pothole formation from longitudinal and transverse crack using mechanistic model. *Journal of Multidisciplinary Engineering Science and Technology*, *8*(11). https://www.jmest.org/wp-content/uploads/JMESTN42353869.pdf

V7. (2022, May). Mean average precision (mAP) explained: Everything you need to know. https://www.v7labs.com/blog/mean-average-precision

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, *30*, 5998–6008. https://arxiv.org/abs/1706.03762

Yao, G., Zhu, S., Zhang, L., & Qi, M. (2024). HP-YOLOv8: High-precision small object detection algorithm for remote sensing images. *Sensors*, *24*(15), 4858. https://doi.org/10.3390/s24154858

Zhang, H., Wang, Y., Dayoub, F., & Sünderhauf, N. (2021). VarifocalNet: An IoU-aware dense object detector. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8514–8523. https://doi.org/10.1109/CVPR46437.2021.00841

Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., & Chen, J. (2023). Detrs beat yolos on real-time object detection. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 16965–16974. https://doi.org/10.48550/arXiv.2304.08069

Zheng, Z., Wang, P., Liu, W., Li, J., Ye, R., & Ren, D. (2020). Distance-IoU loss: Faster and better learning for bounding box regression. *Proceedings of the AAAI Conference on Artificial Intelligence*, *34*(07), 12993–13000. https://doi.org/10.1609/aaai.v34i07.6999